

Notation Summary

Mohammad Sadegh Talebi
University of Copenhagen
sadegh.talebi@di.ku.dk

This note summarizes the most important notations used in the lecture slides by Sadegh and Christian. Christian's slides mostly follow the notations and conventions used in the book by Sutton and Barto [1]. In contrast, those used in Sadegh's material are inspired by notations and conventions of Puterman [2] and of recent literature on reinforcement learning.

The following table offers a mapping between the two notations:

Variable Name	Sadegh's Notation	Christian's Notation
state-space	$\mathcal{S}$	$\mathcal{S}$
action-space	$\mathcal{A}$	$\mathcal{A}$
next-state probability under (s, a)	$P(s' s, a)$	$P_{ss'}^a$
reward function under (s, a)	$R(s, a)$	R_s^a
reward function under (s, a, s')	$R(s, a, s')$	$R_{ss'}^a$
reward model relations	$R(s, a) = \sum_{s'} P(s' s, a) R(s, a, s')$	$R_s^a = \sum_{s'} P_{ss'}^a R_{ss'}^a$
next-state under (s_t, a_t)	$s_{t+1} \sim P(\cdot s_t, a_t)$	$s_{t+1} \sim P_{s_t}^{a_t}$
reward under (s_t, a_t)	$r_t \sim R(s_t, a_t)$	$r_{t+1} \sim R_{s_t}^{a_t}$ or $r_{t+1} \sim R_{s_t s_{t+1}}^{a_t}$
probability of a at s under randomized policy π	$\pi(a s)$	$\pi(s, a)$
expectation over trajectories generated by π	$\mathbb{E}^\pi[\cdot]$	$\mathbb{E}_\pi\{\cdot\}$
value function of π	$V^\pi(s) = \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t \mid s_1 = s \right]$	$V^\pi(s) = \mathbb{E}_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s \right\}$

Another important note is the slight difference between the two considered interaction protocols, as indicated in the figure below.

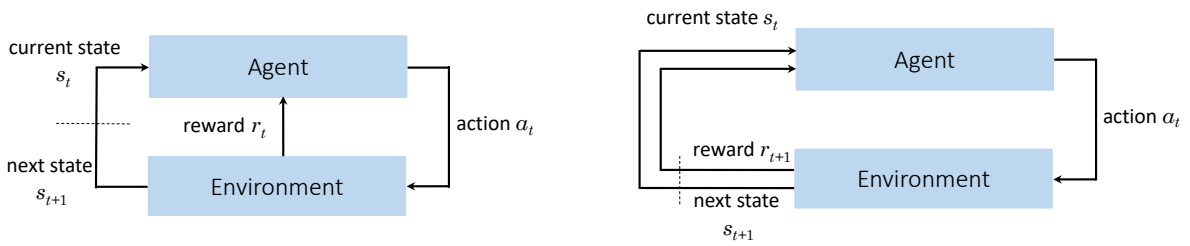


Figure 1: Two interaction models between the agent and the environment (MDP): Sadegh's (left) vs. Christian's (right).

References

- [1] Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.
- [2] Martin L Puterman. *Markov decision processes: Discrete stochastic dynamic programming*. John Wiley & Sons, 2014.